

LOCAL CURRENCY FINANCING

Mitigation strategies for currency mismatch risk in frontier markets

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ABBREVIATIONS

ADB – Asian Development Bank

AfBD – African Development Bank

BIS – Bank of International Settlements

DFI – Development Finance Institution

DMS – Debt Management Strategy

EBRD – European Bank for Reconstruction and Development

EME – Emerging Market Economies

IBRD – International Bank for Reconstruction and Development

ICMA – International Capital Markets Association

IFC – International Finance Corporation

IFI – International Finance Institutions

IMF – International Monetary Fund

LCF – Local Currency Financing

EXECUTIVE SUMMARY

While foreign currency is important and required for debt repayments, import payments and to protect domestic currencies against chaotic macro events, there are persistent currency mismatches that frontier African economies face when borrowing in foreign currency. This is especially so in times of volatility. This over-reliance on foreign currency bring about significant currency mismatch risk. That leads to debt defaults, constrained macroeconomic policy and economic crises.

Local currency financing (LCF) is therefore floated as a strong mediator of these problems, as it ensures debt repayments are made in the same currency as the lending amounts, enhances economic resilience to global shocks, and contributes to local financial markets' deepening. However, despites these benefits, most lenders face significant hurdles to access LCF including high floatation costs, internal biases from the lenders, and insufficient domestic capital. This paper will mostly focus on assuaging these fears of local currency financing by outlining the financial instruments that could be deployed along with guidelines for country-level debt management practices and policy reform. Further, consideration is given to the technological enablers that will be fundamental in the next decades to bolster trust in local domestic markets and promote financial robustness.

PART I: INTRODUCTION

AFRICA'S HISTORICAL RELIANCE ON FOREIGN DEBT

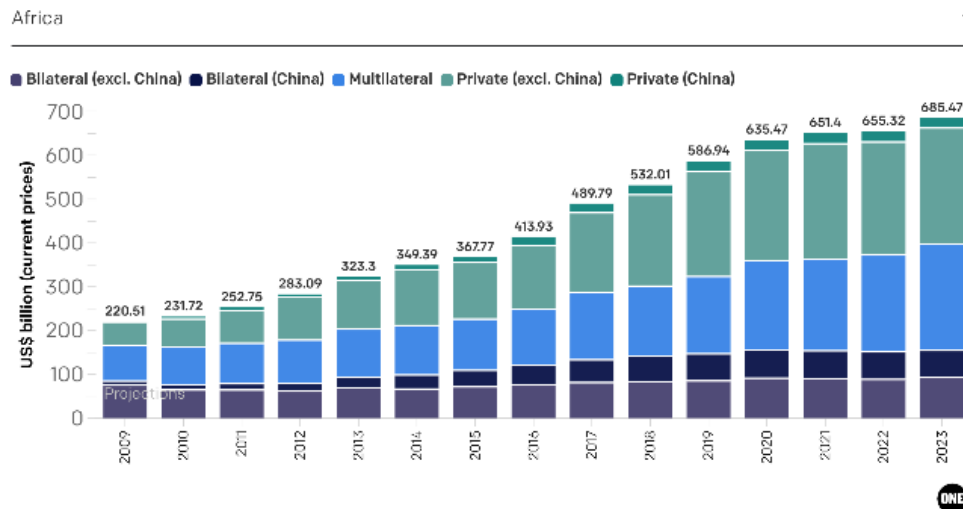
In recent years, African governments have increased their reliance on foreign debt to fund critical projects, exacerbated by the global financial crisis of 2008, the global pandemic of 2020, and the war in Ukraine and conflict in Palestine which slowed the growth potential in the African frontier region (Afrexim Research, 2025). The IMF notes that “since 2010, foreign currency denominated bonds have been the fastest growing source of financing for frontier low-income countries, mainly in sub-Saharan Africa” and that “Eurobond issuances have almost tripled since 2012” (IMF February 2020). As of 2023, external debt reached USD 1.2 trillion which represented about 60% of the region's total public debt stock. From 2008 to 2023, external debt increased from 18.8% of GDP to 41.6% of GDP (Afreximbank Research, 2024). However, based on IMF forecasts, the debt in percent of GDP is set to decline from about 66.8% in 2024 to around 63.5% in 2028 due to strong macro economic predictions around that same period as depicted in the figure below.

Table 1: Macroeconomic Outlook, African Countries, 2022-28

	2022	2023	2024	2025*	2026*	2027*	2028*	Average 2024-28
Growth (percent)	4.5	3.3	4.0	4.4	4.4	4.6	4.6	4.4
Inflation, GDP deflator (percent)	11.7	19.8	17.3	11.2	8.7	7.6	7.2	10.4
Primary deficit (percent of GDP)	-1.5	-1.2	-0.9	-0.4	-0.2	-0.3	0.2	-0.3
Interest rate (percent)	6.0	7.0	8.0	9.0	8.0	8.0	8.0	8.2

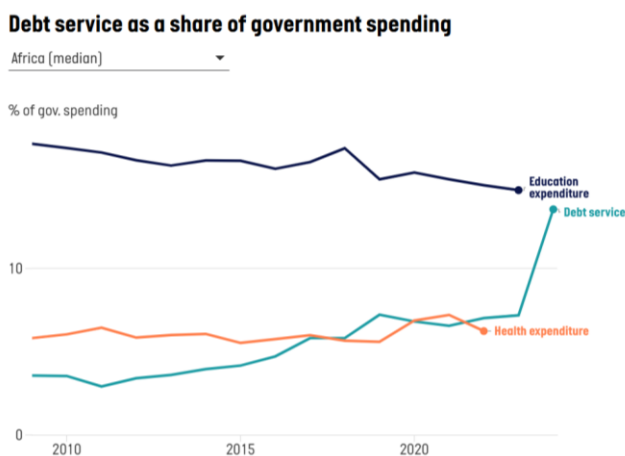
Source: Afreximbank Research based on IMF World Economic Outlook. Note: *Forecast.

Total external debt owed by African countries



Source:

International Debt Statistics (IDS) database.



Source: International Debt

Statistics (IDS), UNESCO Institute for Statistics, WHO GHED, IMF WEO.

The irony, however, lies in that debt service is increasingly taking up the largest use of public funds in greater proportion to what it was intentionally borrowed for i.e. for critical functions in healthcare, physical and digital infrastructure, and education.

Because most of these funds are western nation backed, they are often denominated (about 80-90%) in Euros, Pounds, or Dollars, what is colloquially termed hard currency.

INTRODUCTORY EXPLAINER ON CURRENCY MISMATCH

However, for borrower countries, this kind of financing has had, and continues to create problems with regard to interest and principal payments. This is because local currencies face a heightened level of volatility causing them to fluctuate in value making debt repayment in hard currencies difficult and expensive. In 2023, more than 50% of African countries experienced

currency depreciation, with some currencies such as the Egyptian pound, Ghana cedi, and Sierra Leonean Leone posting nearly 40% negative change in value (Fliss, 2024). Of the 28 countries recording currency depreciation in that period, 17 of them (more than 60 percent) reversed their losses or recorded slowdowns in rates of depreciation (African Development Bank, 2025). These frequent and substantial fluctuations as described by Eichengreen et al. (2002) on the balance sheet expose a country to the risk of local currency depreciation. To compound matters, the terms and condition in which these currencies are signed under stipulate that any currency risks arising from the mismatch is borne by the borrowing party (OECD, Working Paper 117).

AN INTRODUCTION TO THE ORIGINAL SIN HYPOTHESIS

(Hausmann, 1999), claims that this kind of debt reliance, under clearly unfair terms, has been heavily engineered and systemized, terming it the ‘original sin’ - a situation where peripheral or frontier economies are unable to borrow funds in their own domestic currencies. Their key conclusion was that financial fragility was unavoidable in such borrowing-lending systems. All domestic investments either have a currency mismatch (projects that generate revenues in KES will be financed with USD) or a maturity mismatch (long-term projects will be financed by short-term loans). “If a company borrows in dollars to finance a project that generates pesos, peso devaluation can thrust it into bankruptcy. If instead it finances its long-term projects with short-term domestic-currency loans, it will go bankrupt if interest rates rise and credits are not renewed. Unavoidably, investments will suffer from a currency mismatch (because projects that generate local currency are financed with dollars) or a maturity mismatch (because long-term investments are financed with short-term loans).”

Despite, as Han et al notes, the steady growth in foreign investments in frontier markets’ local currency bonds (Bada Han, 2024), since the mid-2000s, -[indeed, one of their conclusions was to negate the original sin hypothesis by proving that EMEs with well-developed capital markets, high levels of institutional quality, and sound monetary policy records are capable of borrowing abroad in safe forms without having to rely solely on foreign currency debt]- there is far from a complete elimination of these original effects, with most lenders preferring to finance in dollars or euros, due first to their own restrictive founding charters and second due to a fear of local currency volatility (Bonizzi, 2024).

PART II. THE PROBLEM

THE ORIGINAL SIN HYPOTHESIS AND ITS EFFECT ON AFRICAN ECONOMIES

In a follow up to their initial paper, Eichengreen et. al. (2003), listed the consequences for countries that face original sin, chief among them being a country’s loss of creditworthiness stemming from real exchange rate volatility. Additionally, wealth effects limit monetary policy interventions. According to Friedman (1957) permanent changes in household resources result in higher consumption ‘during transitory changes leaving spending little changed (Cooper, 2016) Thus, when central banks engage in expansionary measures, these measures may weaken the domestic exchange rate, causing a reduction in net worth and make the policies ineffective in stimulating the economy. In some cases, these expansionary policies go on to behave contractionary despite their intended expansionary design. This renders central banks less willing to let the exchange rate move, and they respond by holding more reserves and aggressively

intervening in the foreign exchange market or adjusting short-term interest rates. The aggregate effects of these wealth effects render these countries unable to achieve sustainable economic stability as their incomes and capital flows are more variable and volatile compared to countries than can borrow in their own currency. This vicious cycle is further perpetuated when international loan markets punish these countries for their inherent financial fragility by imposing a risk premium on development loans advanced.

Moreover, countries that suffer from original sin are less likely to have a floating exchange rate (Eichengreen, 2005). This means that the currency is driven by market conditions with central bank intervening mostly to limit short term exchange fluctuations. This allows a country to maintain an independent monetary policy, giving more leeway to better adjust to external shocks (Stone, 2008). Under fixed/pegged exchange rate systems, monetary policy cannot be used for stabilization purposes. As (Obstfeld, 1995) puts it comprehensively, if a country is hit by a sudden and unexpected fall in the demand of its exports, the harm will be magnified if it uses a fixed system because the central bank cannot intervene to maintain stability. This challenge is particularly pronounced for African economies, where even robust financial and monetary integration alone are insufficient to overcome original sin (Peist, 2023).

HOW CURRENCY MISMATCHES OCCUR AND THE CONSEQUENCES

When such countries accumulate net debt, as developing countries are wont to do, there will exist an aggregate currency mismatch on their balance sheets. The BIS describes it as occurring between domestic and foreign currencies when the borrowing entity's position is sensitive to changes in the exchange rate, the greater such sensitivity, the larger the currency mismatch (Kuruc Emese, B., & P., 2016).

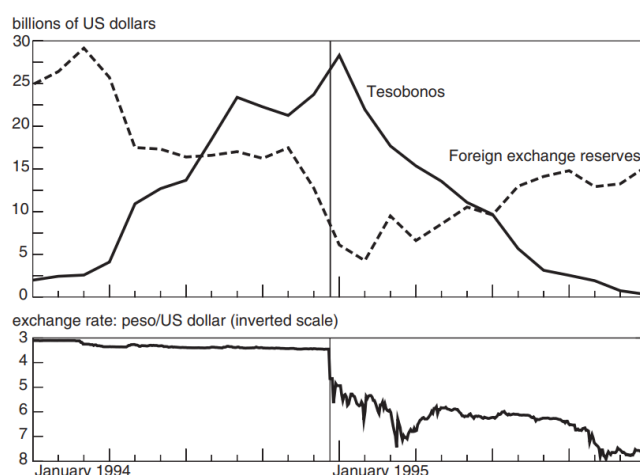
A currency balance sheet mismatch arises if the currency denomination of a borrower's liabilities differs from that of their assets. Movements in the two currencies then affect the valuations of assets and liabilities in all the remaining maturities, affecting solvency. Even without a balance sheet mismatch, the borrower may have a currency maturity mismatch. This is a mismatch between the currency liabilities against the currency assets readily available from foreign exchange income, hedges, or liquid currency. Full hedging only occurs if there is complete absence of currency mismatches affecting valuations over long horizons, as well as in the absence of liquidity shortfalls at shorter maturities.

The dire consequences of these mismatches was aptly put in Goldstein and Turner's book (Turner, 2004) as they described their role in being catalyzers of financial crises, using Asian and Mexican borrowing data to show that large imbalances in their debt repayment plans had high correlations with the start of their respective crises (Morris Goldstein, 2004). This happens due to the interaction between liability dollarization and depreciation leading to dramatic declines in creditworthiness, fall in spending and output, eventually leading to a fiscal crisis. Interestingly, after the crisis broke, these mismatches were sharply reduced, further demonstrating the high correlation.

In this example, of the Mexican peso crisis of '94-'95, the dollar value of Mexican public sector debt, exceeded the rapidly dwindling stock of international reserves by about USD 10 billion. Public sector debt service relative to exports was also significantly higher. Between December 93

and December 94, foreign currency denominated liabilities went from eighty-nine billion pesos to 174 billion pesos, and in 1995, Mexico suffered a 7% fall in GDP, its deepest recession in 50 years.

Figure 2.1 The Mexican peso crisis, 1994–95



More astutely, Goldstein, found another cost of currency mismatch and that is, the cost of getting out of a crisis, with regards to the complications of creating effective monetary policy. They argue that neither contractionary nor expansionary monetary policy would do in such a situation, further constraining central banks' ability to navigate out of the crisis.

Their final observation was that currency mismatches severely handicapped the operation of floating rate currency regimes in emerging markets. Basing this on Calvo and Reinhart's (2000) work that found that developing markets with floating currencies engage in more exchange rate and interest rate market intervention than developed markets, meaning in times of large currency mismatches, they were reluctant to float their currencies and countenance a depreciation of currency, in fear of driving their firms into insolvency.

CASE STUDY: GHANA'S DEBT DEFAULT

To illustrate this problem, Ghana's recent woes with debt repayment come to mind. The country defaulted on its external public debt in December 2022, due to its reliance on foreign currency borrowing, with close to half of their public debt in the previous year being foreign denominated. This debt was covered by revenue streams from its major commodity exports of gold, cocoa and most recently, oil. But when global commodity prices fall as they did in the 80s and 90s as well as in the mid 2010s, the value of the cedi against the dollar also fell, at some points even by 50%. (Zubairu, 2024). Global commodity prices fell at the start of the 1980s, rapidly increasing the size of foreign debt payments which could only be paid out of foreign earnings such as exports. In Ghana's case, the burden of foreign exchange risk was theirs to bear. When the cedi depreciated sharply in 2022 due to external reasons, the country struggled to honour its debt obligations (Bonizzi, et al. 2024). In the same time frame, Ghana's external debt grew from USD 14.7 billion in 2013 to USD 21.1 billion in 2016 (an increase of 44%), yet because of the depreciation external debt went up from 30% of GDP in 2013 to an about 56% in 2016 (an increase of 87%). (Jones, 2019). The punishment cycle began as most investors

throughout the year 2022, consistently required a higher return on investment for holding Ghana's debt compared to the safer U.S. Treasuries. One of the first solutions offered to its debt holders was a DDEP and debt exchanges for domestic debt denominated in US dollars and Cocobills issued by the Ghana Cocoa Board. Further, credit default swaps were issued as insurance coverage for the country's sovereign debt. (David, 2025). (Osei, 2000) argue that the critical factor that caused Ghana's debt default was the country's inability to generate sufficient foreign exchange through export earnings leading to a current account deficit. Second, is the low rate of return on investment to which borrowed funds are applied i.e. much of the capital inflows went into infrastructural investments that have long gestation periods and/or large indivisibilities, so that returns are low, especially in their early years. And the third factor is the low domestic saving rate.

CASE STUDY 2: KENYA'S SGR RAILWAY

For large scale infrastructure projects, both local and foreign debt is needed. This is because large scale projects will need access to imported construction material and therefore funding in hard currency. Nevertheless, these projects can still benefit from becoming more financially resilient when the incomes they generate match their financing. A good example is the Kenya SGR railway which was financed in local/foreign debt. Since most if not all the income is generated by Kenyans using the train for intercity travel, it would be a massive help to pay back in local currency. (Ndii, 2020) explained this much more clearly by breaking down the numbers to show the dire situation that the SGR project had placed Kenyan taxpayers. The initial outlay was USD 3 billion mostly financed by Chinese loans that had requirements for repayment of both principal and interest, necessitating a higher revenue burden. Annual costs amounted to KES 50 billion with all revenues denominated in the Kenya shilling which at that point depreciated 18 percent since T0, raising the interest cost by KES 3 billion a year. The major issue is that the bulk of its approved tariffs and direct costs are in foreign currency. The best alternative is trucking which has less foreign currency exposure and indirect costs. Depreciation of the shilling would lead to an erosion of the railway's cost advantage over trucking.

To resolve this situation, part or all of the funding package should be structured as a local currency project bond rather than hard currency direct loans. An additional provision of guarantee by a development finance partner would help local institutional investors take a first step towards financing their own local infrastructure through simple instruments.

PART III: THE STRATEGIC ROLE OF LCF

RATIONALE FOR LCF

First, is the previously discussed role that currency mismatches have on all of the financial crises in emerging markets. While developed economy borrowers may be able to borrow funds internationally in their own domestic markets, borrowers from emerging markets are often unable to do so due to insufficient local financial resources available, tenors in the local market are too short, or borrowing offshore offers lower interest rates.

Multi-development banks (MDBs) may take part in local currency financing bestowing benefits such as (Hoschka, 2005) describes:

1. Deepening local capital markets – by following best-practice issue standards, setting new benchmarks, providing role model transactions in terms of documentation and execution, stretching the yield curve, introducing innovations with respect to available financial instruments in local capital markets, and providing significant diversification opportunities for local institutional investors such as insurance companies and pension funds.
2. Development of highly liquid derivative markets – which is key to reduce tenor and currency mismatches of local borrowers.
3. Extending broader macroeconomic benefits – as argued previously, currency mismatches exacerbate financial crises in emerging markets but by strengthening the local financial system, the need to borrow foreign funds will be reduced. Further, by issuing in local currency, an internationalization of the local currency will occur, attracting foreign investors, thus allowing local issuers to raise capital internationally in local rather than in foreign currency.

ADVANTAGES OF LCF FOR DIFFERENT STAKEHOLDERS

For SMEs in frontier markets, LCF is helpful because the income used to make repayments will be in the same currency as the financing they received. This is particularly useful for small businesses as often their proceeds are in the form of local currency. Any financing they receive in foreign amounts is a boost to their operations but becomes a burden to pay when their trading is limited to local and regional customers.

Although large scale infrastructure projects may require a blended form of finance due to the necessary imports of inputs, they can still benefit from local currency finances to help in setting labour wages and also helps to offset the revenues that will most likely occur in local currency.

Further, LCF reduces an economy's dependency on foreign currency, enabling resilience to global financial shocks thus promoting long-term economic development and poverty alleviation (Paul Horrocks C. M., 2025). This includes adverse economic shocks from currency depreciation, making repayment obligations manageable and predictable. This leads, eventually, to the development of domestic capital markets for bonds and currency swaps with local currencies. There is therefore a greater ability to predict the efficient use of domestic funds, freeing up the national budgets for other investments.

CASE STUDIES OF SUCCESSFUL LCF USES

1. The African Development Bank (AfDB) has been loaning out in South African rand since 1997 through the cross-currency swap market and by issuing rand bonds in the offshore Euromarkets, not just to public and private sector borrowers in South Africa but also to public sector borrowers in the Southern African Development Community.
2. The IFC provides local currency debt financing in three ways: (i) loans from IFC denominated in local currency; (ii) risk management swaps that allow clients to hedge existing or new foreign currency denominated liabilities back into local currency; and (iii)

structured finance solutions such as partial credit guarantees that enable clients to borrow from other sources and risk sharing facilities, and to participate in securitizations. In Mexico, a USD9.6 million equivalent 9-year bond issue from the IFC's Municipal Fund, with the IFC partially guaranteeing the bonds which raised the credit rating IFC partially guaranteed the bonds. In Johannesburg, the IFC provided a partial credit guarantee for a USD150 million equivalent 12-year issue, enhancing the credit rating of the bonds from A- to AA-, and allowing the extension of the maturity of the city's borrowings from 6 to 12 years. What this means is that IFC takes on the refinancing risk, ensuring that any obligation by the city in case of a call on guarantee is in rands not dollars.

Other examples include IFC's role in the first securitization by Sonatel in Senegal, which at the time was West Africa's first-ever asset-backed security, and which helped act as a derisking signal to mobilise private capital from local and regional investors; the first sub-Saharan African gender bond issued by Tanzania's NMB Bank, and a green bond issued by Redefine Properties Ltd in South Africa where IFC's role as an anchor investor was critical to the success of the bond issue, helping to catalyze an additional 750 million rand from other investors. IFC's support also enabled the company to raise local currency funds with tenors of seven and ten years, helping to shape the local market which has largely been characterized by shorter tenors of three and five years. (Monnier)

3. Similarly in the Philippines, a cross-currency swap exists where ADB will swap a given amount of foreign currency in exchange for the equivalent in local currency. At the termination date of the swap, the transaction will be unwound and ADB will repay the local currency in exchange for dollars. During the life of the swap, ADB will lend the local currency to creditworthy financial institutions in the country which, in turn, on lend to developmentally sound investments and projects that require long-term local currency financing and meet ADB's regulations on environment, resettlement, and Indigenous people. Thus, the ADB will absorb the risk of the country and the participating banks, both domestic and foreign. The participating banks will in turn absorb the commercial risk. In this manner, each player will take on risks that it is best equipped to absorb.
4. The EBRD sources local currency for financing through single currency revolving facilities, cross-currency swaps, and local currency bonds issued domestically or in the Euro markets. Mostly loans are lent out to sub sovereign borrowers such as city governments and municipal entities particularly for infrastructure projects for example a RUB346 million (ten million Euro equivalent) loan to Russian Archangelsk municipal water utility and a PLN108 million (24.5 Euro million equivalent) to the Bydgoszcz city-owned municipal water and sewerage company.

PART IV: OBSTACLES TO LCF ADOPTION

CHALLENGES TO BORROWER COUNTRIES

Developing economies may not be able to use foreign exchange derivatives instruments in the management of their public debt portfolios due to the prohibitive costs associated with accessing global derivatives markets —mainly owing to insufficiently established or poor credit ratings. (OECD, Work Paper 117). Financial factors such as interest rate differentials, international market conditions and short-term financial flows also influence the likelihood of sudden and

large exchange rate movements (World Bank, 1987[4]), make it difficult for low-income borrowers reliant on predictable FX rates to manage their hard currency debts.

Second, and perhaps an even more primary challenge is the lack of sufficient domestic capital that borrower countries have to contend with. This is especially so in comparison to the scale and number of projects that have to be funded by these savings. This gap necessitates them to rely heavily on external financing, that is set at the foreign lender's terms and saddles the borrower nation's populace with generational debt. (Barden, 2025).

Lastly, the heightened volatility that frontier and emerging currencies undergo that causes their value to have sharp fluctuations in value adds another level of risk to lend to them as well as makes it expensive and difficult for them to make repayments.

CHALLENGES FOR THE IFIS AND MDBS

When a currency mismatch exists between the funding currency and the currency in which the capital is deployed, significant risk is introduced. IFIs need to ensure the returns generated by their assets allow them to cover their operating costs. Deploying capital and generating income in a different currency therefore risks them not being able to meet these costs. One of the biggest reasons they do this is the need to repay the investors they borrow from in hard currency, increasing their need for hard currency assets and thus the motivation for hedging against any open exposure to currencies different from those the debt is issued. Further, the net interest margin resulting from borrowing in US dollars or Euros is a revenue driver for IFIs that would be complicated to replicate in local currency pools where the premia paid to lenders may be difficult to pass on to borrowers.

In relation to sovereign lending (i.e. loans provided by development finance institutions or MDBs directly to national governments), the ability to provide concessional funding is dependent on accessing these same deep and liquid pools which are often in hard currency.

Further, in developing markets where the bulk of institutional investors' portfolio is concentrated in government securities, such as in Kenya where pension funds and insurance companies allocate 47.8% and 78% respectively to government bonds, diversification towards infrastructure or SME financing in local currencies would lead to critical vulnerabilities in the stability of government financing (Paul Horrocks C. M., 2025).

For these reasons, most IFIs and MDBs have prerequisites that they would need met before issuing out LCF, according to (Hoschka, 2005):

1. That the MDB has a tax-exempt status in the borrowing country
2. That it is free from additional domestic rating.
3. That there is broad investor access from insurance companies and pensions, etc, in order to lower funding costs.
4. That the risk weightings of MDB bonds be close to 0% as recommended by the Bank for International Settlements.
5. That MDB bonds be exempted from statutory reserve and liquidity requirements in order to improve liquidity and broaden distribution.

BEAC CHEVRON CASE

A developing case is around the custodianship of oil remediation funds which companies keep for funding cleanup and remediation of oil waste after their drilling activities. These funds have traditionally been held in US or European banks and consequently most oil companies have gotten used to this system. This clashes with the Bank of Central African States' and the member states own wishes to have the funds (that are usually set up in American or European escrow accounts) repatriated and stored in members states accounts, in this case under BEAC's custodianship. They wish to use these funds to protect the CFA franc from volatility by shoring up reserves and also using them as collateral for debt restructuring. Post-Covid, the six member states struggled with low foreign currency reserves to meet important debt payments of course the oil companies backed by us government have raised protestations citing the need to have the funds remain untouched, and inaccessible by the governments for any other reasons than remediation. Although this is not a strict financing case, it will still be used to highlight the importance of local financing structures to avoid defaults, eliminate foreign currency dependency and strengthen domestic financial markets. The BEAC case described above is a solid example of the desperation that can result from foreign currency dependency and the ripple effects of poverty alleviation.

PART V: MECHANISMS FOR MITIGATING CURRENCY RISK

INNOVATIVE FINANCIAL INSTRUMENTS

BERMAN'S SYNTHETIC LCF BOND

Berman (2020) makes the point that hard currency debt is, often, not the right kind of debt for borrowers in LDIC's. The solution proffered is to transfer foreign exchange risk to international investors' balance sheets as they will typically have the capabilities and appetite to shoulder such risk.

Essentially, Berman suggests that when the IFI loans out to the borrowing country in local currency, it will then create a currency swap with a hedging provider who will collect currency risks from many different countries, package them and sell them to investors with high-risk appetites via a synthetic local currency bond. By buying up these bonds, the investors take on the currency risk fears away from the original lender and pass them on to investors who are specialized to profit from such volatility.

This is a unique and innovative asset class for investors, with the double effect of allowing IFIs to finance themselves in hard currency at a lower cost and increasing the hedging provider's capital efficiency; and deepens the capital markets of the borrower country.

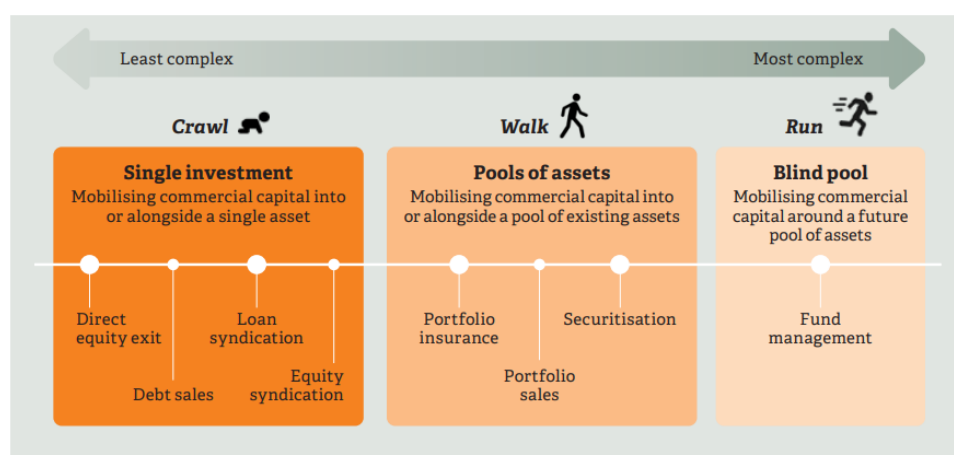
The effectiveness of such a bond could be enhanced by standardizing across the different economic regions, which could help cut on the flotation costs, and by having a unified

framework on rating methodology, minimum pool sizes as well as automated pricing, hedging and settlement systems.

SECURITIZATION TECHNIQUES

Enabling risk transfer to currency through securitisation involves pooling financial assets to create marketable securities. An example technique is the originate-to-share model. In this model, capital exposure is distributed or delegated to commercial investors instead of being tied up to its original provider until maturity, otherwise known as the traditional buy to hold model (Wasim Tahir, 2023). It works in the following way:

1. Simply, by selling it in the secondary market especially if the project has passed its early stages and presents a clearer credit risk (such as once an infrastructure construction phase is passed and the project begins operations) or by acting as lead anchor in a loan syndication where an IFI negotiates terms of the loan but contributes only a fraction of the funding, with the bulk of it provided by other lenders; or
2. By taking up an insurance cover that covers an IFI's credit losses from a pool of assets transferring the risk to the balance sheet of the insurer. Alternatively, by issuing securities whose value varies with the returns on a specified pool of assets held by the IFI so that risk is transferred to investors in the securities; or
3. By attracting commercial capital into a "blind pool" of assets



The originate to share model, as described from simple to sophisticated. (Wasim Tahir, 2023)

The *Singapore's Bayfront 3* has also been successful in securitization by building a platform that provides structured financing mechanisms to facilitate the pooling and securitisation of DFI and MDB loans. It does this by purchasing infrastructure debts from financial institutions and distributing them to institutional investors through securitization. Some of the most applicable projects include a USUSD401 million securitization of infrastructure loans that had a sustainability tranche, proceeds of which will be used to finance and/or refinance the purchase of loans for renewable energy and affordable social infrastructure projects. The process flow is as follows:

1. BIM will acquire predominantly brownfield project and infrastructure loans from financial institutions, warehouse and manage them, before securitizing them and issuing structured notes to investors

2. The platform will connect project owners, institutional investors, and partners to mobilise capital, fund higher credit volumes and recycle capital at scale.
3. This approach frees up IFIs' balance sheets and gives investors exposure to a new asset class.

How the Bayfront model could work for African frontier economies is to have a pipeline that purchase local currency infrastructure loans from regional governments and financial institutions, pool them and securitize them into tradeable instruments for pension funds, insurance companies and other institutional investors to build investor confidence, creating new financial products and spreading risk to investors with the appetite for it. For it to work, secondary market liquidity buoyed by the presence of market makers is crucial.

FIRST-LOSS GUARANTEE

A guarantee is a promise by one person to take responsibility for another person's obligations if the latter defaults or fails to perform on her obligations, e.g. a failure to meet loan repayments or to redeem bonds, or expropriation of an equity stake. That is, a guarantee backstops payments, typically covering political and/or commercial (e.g. credit, regulatory/contractual) risks that investors are unwilling or unable to bear (Halvorson-Quevedo and Mirabile, 2014).

A first-loss guarantee is a type of guarantee in which the guarantee provider agrees to bear losses incurred up to an agreed percentage in the event of default by the borrower. Its purpose is to reduce risk and attract lenders and investors who may be hesitant to participate in a financing arrangement due to risk concerns. In a first-loss guarantee, a guarantor (typically a DFI or multilateral institution) absorbs initial losses up to a predetermined threshold, reducing perceived risks and enhancing the attractiveness of investments in a volatile market (World Bank Group, 2024).

Prominent examples include: MIGA's Infrastructure Guarantees which promote foreign direct investment into developing countries to support economic growth through various guarantee structures and Africa Energy Guarantee Facility which provides credit enhancement specifically for energy infrastructure projects across the continent. They will typically cover 10-20% of total project value and automatically activate when a project's returns fall below a predetermined threshold. The guarantees cover against sovereign or sub-sovereign non-payment in a power purchase agreement, expropriation and breach of contract, currency inconvertibility, war, civil unrest and arbitration award default. (African Energy Guarantee Facility, n.d.). By reducing perceived financial risks, guarantees widen the ability of more institutional investors to participate.

Other facilities include the T&C facility offered by the EDFI Management Company which helps cover transferability and convertibility (T&C) risks for renewable energy projects located in countries facing significant foreign exchange reserve constraints. Transfer risk is the risk of inability of a debtor in a given country to meet their financial obligations due to lack of availability of the currency in which the obligation is denominated regardless of the debtor's particular financial condition, while convertibility risk reflects the scarcity of foreign currency in host countries in times of foreign currency stress. The T&C cover mitigates the risks of inconvertibility and non-transferability between local currencies and hard currencies, supporting

DFIs' high-risk investments (European Commission, 2021). This provides breathing room for any evolving currency crises to resolve while maintaining project viability and offering cover for up to 12 months of principal and interest payments over a maximum of seven years. One of the most pertinent benefits of the T&C Facility is in the energy sector usefulness to the renewable energy sector because electricity is sold in local currency while power plants are typically financed in US dollars or euros. By having a reserve account for each project, the T&C facility enables more efficient and effective private financing of renewable energy projects across low-income countries.

INFRASTRUCTURE GUARANTEES

Infrastructure investments are characterized by significant upfront costs and their particular risks can be grouped in three broad categories: political, macroeconomic and technical risks. (Introductory Guide to Infrastructure Guarantee Products from MDBs, 2018).

Infrastructure guarantees mitigate these risks associated with investing in infrastructure projects, enabling investors to mobilise local and hard currency funding and enhancing the creditworthiness of these investments.

Further, given that any infrastructure project will be associated with at least some local currency portion and will therefore need some local currency funding, blended financing could be an appropriate approach for infrastructure bonds. The typical structure look like:

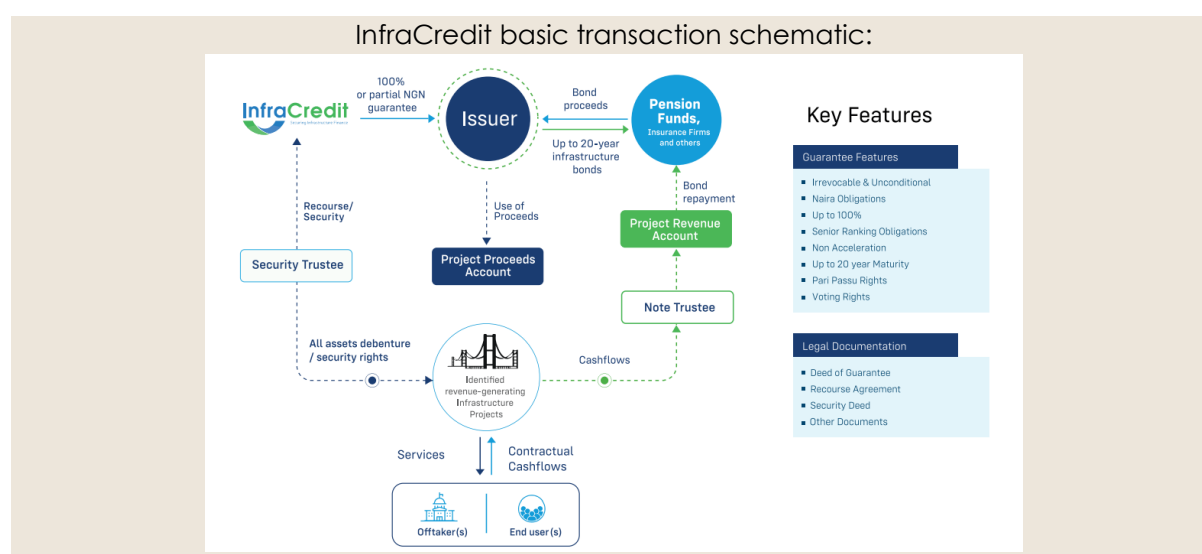
1. A Concessional Layer: 10-30% from DFIs at below-market rate
2. The Commercial Layer: 70-90% from private investors at market rates
3. A Guarantee Layer: Credit enhancement from multilateral institution

This would be highly useful for projects in transport, energy and water infrastructure as the local currency financing could come in the construction phase: to cater for local labour and materials; in the operations phase: catering for local staff, maintenance and utilities and in revenue generation: so that tariffs and fees are collected in local currency. It could also work for climate and environmental financing as in the case of the Global Environment Facility who found in their report that by providing a total of \$1.4 billion in climate finance (as defined by OECD-DAC's "Rio Marker"), equivalent to 68% of total funding from the GEF from 2013 – 2014. that period, 60% of which was from the private sector. This finance mobilized about \$800 million from the private sector, implying that GEF climate finance, on average, mobilized about sixty cents from the private sector for each dollar from the GEF. Similarly, from mobilizing \$175 million of GEF funds for blended finance operations mobilized, they were able to draw in \$1,098 million from the private sector, equivalent to a leverage ratio of 6.3. their experience proved that Blended Finance can be an effective instrument to redirect private sector finance towards investments that benefit the sector critical industries. (GEF)

A particular example is Nigeria's Infracredit which provides local currency guarantees to enhance the quality of debt instruments issued to finance infrastructure projects in Nigeria, thus attracting pension funds, insurance firms and other long-term investors to invest in long-term bonds for infrastructure projects in Nigeria. The provided guarantees are irrevocable and unconditional. So far, they have closed over twenty-one infrastructure projects and guaranteed a total of 474

million USD in debt and have mobilised 118.7 billion naira in domestic capital from local institutional investors.

Their area of coverage cuts across sectors from power and renewable energy, water and waste management, to agriculture and manufacturing, among others. Some of the biggest benefits felt in the Nigerian markets include financial inclusion by ensuring first time issuers have access to the domestic bond markets; financial deepening by extending bond tenors for corporates and broadening the pension fund investor base; and financial innovation by introducing new fixed income instruments such as green bonds (Infracredit Nigeria, 2025).



Similarly, in Argentina a USD 480 million guarantee was issued to the government from the IBRD as a backstop to the government's payment obligations, thus assisting to create a global market and mobilizing about USD 3.2 billion of mostly private investments. By doing so, Argentina was able to launch the Renovar program, to increase the share of renewable energy production to about 20 percent by 2025 (World Bank, 2018).

LIQUIDITY FACILITIES

Structures such as foreign exchange liquidity facilities can be designed to temporarily address the borrower's inability to meet debt servicing obligations resulting from an adverse foreign exchange rate movement. Based on the idea that real exchange rates are less volatile than nominal exchange rates and that it is therefore possible to cushion projects against this more manageable level of volatility with a small facility, if calibration is done correctly.

An example is the IBRD's Catastrophe Deferred Drawdown Option (Cat DDO) which is a contingent credit line that provides immediate liquidity to IBRD member countries in the aftermath of a natural disaster. It gives a government immediate access to funds after a natural disaster, a time when liquidity constraints are usually highest. (PPIAF, 2025).

Such a structure could be altered to not only cater to natural catastrophes but also to activate when the currency devaluates to a set floor; or for commodity reliant countries, if there are

commodity price shocks; or if there is risk of spillover effects from stress factors in neighbouring countries.

HEDGING SOLUTIONS

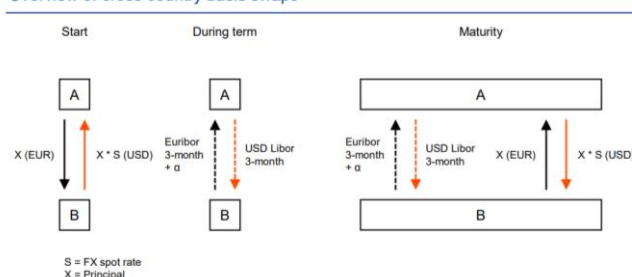
Hedging solutions allow development finance institutions to continue lending in hard currency while ensuring that borrowers receive local currency funding. This is primarily done through instruments such as non-deliverable cross-currency swaps and forward contracts and is especially useful for emerging and frontier market currencies.

Cross currency swaps are used by market participants as a means of hedging currency exposure or speculating on currency direction over a given period of time. A cross-currency swap occurs when two parties simultaneously lend and borrow an equivalent amount of money in two different currencies for a specified period of time. It entails an exchange of interest payments in one currency for interest payments in another. In addition to the exchange of interest payments, there is also an exchange of principals (in the two different currencies) at the beginning of the contract and at the end, at the spot rate prevailing when the swap is initiated. (Brophy, 2019).

The process flows like this:

1. Initial Transaction: DFI lends USD to project, which is immediately swapped to local currency.
2. Periodic Payments: Project pays local currency debt service, converted to USD for DFI
3. Risk Transfer: Hedge provider assumes currency risk in exchange for fees.

Overview of cross country basis swaps



An example would be a 200 km highway under construction financed by 150m USD from a DFI with revenue collection in KES which will use Forward contracts covering construction period plus 10-year operation. This gives the benefit of Predictable cash flows in local currency. Another example is a 100-mw solar plant financed with 80m USD and generates revenue in local Naira using a hedging structure of a Non-deliverable NGN/USD swap covering 7-year loan term. The benefit is that the project avoids currency risk while AfDB maintains USD exposure.

(Hoschka, 2005) writing two decades ago when frontier markets were not as liquid as they are now, stated that cross currency swaps can be structured in smaller amounts and thus can be closely matched with the disbursement schedule of a loan. Additionally, they can be structured with an amortizing schedule that corresponds to the amortization schedule of the loan to be financed. Both of these advantages can eliminate the issue of “negative carry” that may be associated with a local currency bond issue.

A more concrete example is the IBRD's offer for borrowers to convert or swap their foreign currency-denominated IBRD liabilities into local currency. To hedge its risks, IBRD enters into matching currency swaps from market counterparties. However, this arrangement is limited to countries where an active and liquid cross-currency swap market exists.

The IBRD notes that local currency hedging transactions such as described above may not always be ideal for sovereign borrowers but may be ideal for sub sovereign borrowers (counties and municipalities) that conduct their fiscal management independently from the central government.

One developing country that has engaged in such activities is Trinidad and Tobago. The country has considerable external debt, denominated in Japanese yen, which was partially taken over by the central bank. The widely fluctuating foreign reserves (earnings from oil exports) are in U.S. dollars, because the Trinidad and Tobago dollar is pegged to the U.S. dollar. The central bank has used currency swaps to hedge against changes in the U.S. dollar-yen exchange rate, as well as swaps from floating-rate into fixed-rate liabilities. (Catharina, 1997) .

Hungary, as well as other developing CEE countries, also engaged in loan conversions in the mid- 2010s by enacting laws that allowed their citizens with Swiss-dominated liabilities into domestic currency loans. This was in response to the Swiss National Bank (SNB) discontinuing its policy of maintaining the minimum exchange rate of 1.2 Swiss francs against the euro, raising borrower rates up to 20%. (Yeşin, 2022) found that by doing so the exposure of CEE financial institutions to systemic exchange rate risks to their balance sheets through domestic currency depreciations was reduced. CEE banks were heavily exposed to currency risk and loan conversion programs were able to reduce financial stability risks linked to Swiss franc loans in the CEE region.

MARKET DEVELOPMENT AND SUSTAINABLE FINANCING

Scaling the issuance of sustainability-linked bonds with aims of leveraging national development banks not only helps to mobilise local capital, but also provides investors with an innovative method for channelling finance into areas such as biodiversity and climate change adaptation, which can be challenging to finance using green, social and sustainability (GSS) bonds in certain situations due to a lack of pipelines of sizeable and bankable assets that GSS bonds must, by definition, be linked to finance. African Examples: South Africa's Green Bonds: Government issued sustainability bonds linked to renewable energy targets Nigeria's Sovereign Green Bonds: Tied to forest conservation and renewable energy goals Kenya's Infrastructure Bonds: Could be restructured to include sustainability metrics. KPIs for the African context could include 1. Energy Access, for example, the percentage of population with electricity access 2. Renewable Energy: Share of renewable energy in national energy mix 3. Financial Inclusion: Percentage of population with bank accounts 4. Infrastructure Quality: Road density, telecommunications coverage.

To overcome the problems associated with small economic size and small banking systems—including illiquid debt instruments, short maturities, a restricted and undifferentiated investor

base, and undeveloped secondary markets—regional bond market development initiatives such as those initiated in Asia should be promoted in SSA, including initiatives for harmonizing legal and regulatory frameworks and for facilitating the cross-listing of bonds on several national exchanges.(developing local currency bond markets in SSA)

REGULATIONS

The World Bank’s guidelines for developing local currency bond markets (Hashimoto, 2021) provide a framework for assessing the legal and regulatory environment of a market, with a graduated scale from 1 – 5 depending on the depth and comprehensives of the laws to protect investors, handle taxation, and develop both primary and secondary markets.

In the framework, they recommend:

1. A clear and single entity within the borrowing authority
2. Clear roles outlined for the debt management entity.
3. Rules allowing the authority to conduct broad debt management operations for risk management (such as liability management operations)
4. Regular preparations and, publication of a medium-term debt management strategy
5. The prohibition of unfair practices (such as market manipulation, front-running, and collusion)
6. Safeguarding investor ownership rights, such as the segregation and identification of customer assets
7. Standards for the eligibility, governance organization, and operational conduct of a CIS operator
8. Rules governing the legal form and structure of CISs and the segregation and protection of customer assets.
9. Rules for determining the tax treatment of returns from government bonds (such as interest, discount, and principal) for retail, institutional, and non-resident investors)
10. Clear and sound rules for determining the tax treatment of secondary market transactions for government securities.
11. Rules promoting effective and efficient tax alignment with respect to investing or trading in financial instruments between key institutional investor groups.

By following these guidelines, and by investing in greater economic size, larger banking sectors, greater openness to trade, and better regulatory quality and respect to the rule of law, a stronger local bond market can be developed.

For better monitoring of domestic sovereign debt, an appropriate debt management strategy needs to be put in place. It is also important to train personnel in the field of debt management adequately because these skills are needed for issuing domestic securities. (Berensmann, 2015).

STRENGTHENED PUBLIC DEBT MANAGEMENT

It cannot be overstated how important it is for borrower countries to be proactive in effectively structuring their debt portfolios in terms of maturity, currency, and interest rate composition.

The World Bank and IMF (Estevao, 2019) jointly produced a report giving an 8-step framework for developing an effective debt management strategy, which is summarised below:

1. Identify the objectives for public debt management and scope of the DMS.
Describe the overall objectives for debt management and the scope for the DMS.
2. Identify the current DMS and analyse the cost and risk of the existing debt.
Gather detailed information on outstanding debt, including the servicing profile and main portfolio risks.
3. Identify and analyse potential funding sources, including their cost and risk characteristics.
Assess the characteristics, including the constraints, cost, and risk, of all available and potential financing instruments, and how they might mitigate the portfolio risks previously identified.
4. Identify baseline projections and risks in key policy areas—fiscal, monetary, external, and market.
Detail baseline projections for key fiscal, monetary policy, external, and market variables, hashing out a clear and comprehensive set of country-specific risk scenarios to be evaluated.
5. Review key longer-term structural factors.
Articulate the long-run structural factors that the DMS should take into account.
6. Assess and rank alternative strategies on the basis of the cost-risk trade-off.
Rank a small number of candidate strategies in terms of cost and risk.
7. Review implications of candidate debt management strategies with fiscal and monetary policy authorities, and for market conditions.
Ensure that the candidate strategies are consistent with fiscal and monetary policies, while maintaining debt sustainability, and are in line with plans for market development.
8. Submit and secure agreement at elevated levels on the DMS.
Secure the approved DMS.

Further, it is recommended that government officials shift focus ways from short term cost saving measures that often lead to the accumulation of short-term, floating-rate, foreign currency-denominated debt. This will lead to an increase in vulnerability to currency risk during period of high economic upheaval. A critical aspect of this is to effectively measure exchange rate exposure that is highly correlated to interest rates, as well as their high correlation to bond markets.

TECH ENABLERS

CBDcs

Central bank digital currencies (CBDC) are a digital form of cash issued and regulated by central banks and are less volatile than crypto based assets. One of their biggest benefits within the context of LCF is the possibility to streamline cross-currency transactions therefore reducing the need for foreign currency reserves.

Further, they have the potential to have the potential to spur competition in payment markets, increase access to bank accounts, enhance financial inclusion for underserved populations, and modernize outdated legacy technology platforms.

In Africa, Nigeria, Kenya, South Africa and Ghana are at advanced stages of introducing their own versions of CBDCs/ There are risks associated with their implementation for example having a negative impact on bank profitability due to deposit substitution, where citizens might pull money out of commercial banks to hold CBDCs, affecting banks' ability to lend. This is especially so in countries with unstable financial systems. There is also a huge operational burden to manage the systems as well as challenges with regard to data protection.

Yet the advantages far outweigh any risks, as they can promote dynamism in foreign exchange reserves through direct central bank-to-central bank lending; algorithmic systems that dynamically adjust CBDC floats based on trade flows and other economic indicators; and automated liquidity provisions in stressful periods.

DLT AND BLOCKCHAIN

Distributed ledger technology for regional bond settlements. Distributed Ledger Technology (DLT), including blockchain, is identified as a powerful tool for regional bond settlements and enhancing financial market efficiency by facilitating efficient and secure settlement of bonds denominated in local currencies across regions. The Abu Dhabi Securities Exchange (ADX) recently listed the first DLT-based bond in the MENA region using HSBC's digital assets platform, with significant efficiencies in post-trade and asset-servicing processes, streamlining issuance flows, creating unique identifiers, and compressing settlement cycles being promised to be the biggest benefits.

The ICMA are clear that DLT bonds in and of themselves do not reduce settlement risk but can be paired with digital programmed cash solutions such as a CBDC to be exchanged simultaneously on the same DLT network, also known as “atomic settlement,” or intraday settlement. Atomic settlement or shortened settlement cycles could eliminate trade fails associated with incorrect settlement instructions, increase transparency to ensure available inventory, enable real-time netting for positions held on the DLT or blockchain network, and reduce or eliminate counterparty risk. (International Capital Market Association, 2022).

Nevertheless, in a discussion on El Salvador’s bitcoin bond, found that cryptocurrency bonds were not an effective path to pursue, does not enable a sovereign to break away from original sin, and moreover, it does not sufficiently mitigate the risks associated with being unable to borrow in locally-denominated currency the minor ways in which cryptocurrency borrowing mitigates original sin do not make up for the excessive risk, volatility, and lingering concerns associated with borrowing abroad in a foreign—or digital—currency (Adams, 2023). This is a good learning point for frontier markets that are looking to incorporate DLT into their systems to mitigate currency mismatches. They can do this by focusing on operational efficiency rather than prioritizing investment returns, for example, by looking at its capability to streamline settlements, and the ability of smart contracts to enforce developmental outcomes and that allow for real-time compliance reporting to multiple regulatory authorities and investment partners.

PART VI: CONCLUSION AND RECOMMENDATION

This report detailed the importance of local currency financing for African borrower countries and has shown the systemic phenomenon of "Original Sin," has created a vicious cycle of inherent fragility for many of these countries. A case study like Ghana shows how extreme the effects could be, leading to domestic financial crises if external currency shocks are not well managed. LCF is essential for building genuine financial resilience, fostering the depth and liquidity of domestic capital markets, and ultimately achieving sustainable development goals.

While plenty of obstacles exist from insufficient domestic savings, shallow bond markets and high hedging costs as well as lender biases for hard currency, they exist to emphasize the importance of local currency financing. A multi-faceted approach from policy reform, innovative financial instruments, wholistic development of local bond market and responsible adoption of technology enablers.

Altogether this paper has attempted to approach the problem of local currency financing with a view to suggest quickly implementable solutions to the problem as it affects many frontier nations. In doing so, particular attention is given to the breadth of innovative financial instruments with care taken to ensure that the solutions are tailored to African contexts, budgets, and scaling ability. A brief section is set aside to discuss nascent technology of blockchain technology and central bank digital currencies and the role they can play in this situation.

A summary of the recommendations is given below. For borrowing countries, this involves:

1. Fostering the growth of local bond markets by encouraging healthy domestic savings and diverting them into developing deep and liquid local currency bond markets
2. Enhance transparency and governance and the rule of law as they are critical for building investor confidence. As demonstrated by the BEAC case, clear, predictable, and structured financial governance is critical for attracting foreign capital.
3. Strengthen the debt management capacity by investing in training staff and developing technical expertise to craft tight debt portfolios with accurate and measured foreign exchange exposure.

For IFIs, and MDBs other lending institutions:

4. Tightly managing their balance sheet exposure as well as expanding their role as anchor financier in order to de-risk investments for other institutional investors and attract broader participation.
5. Actively promote synthetic LCF bonds, securitization platforms, first-loss guarantees, and foreign exchange liquidity facilities as hedging tools and platforms that transfer currency risk to entities that are well equipped to manage it. This will lower the costs of issuing local currency financing, increasing its availability.
6. Continue to provide technical assistance and capacity build in countries with weak debt management practices in order to understand how to navigate complex financial derivative markets and develop their own domestic bond markets.

More concrete recommendations include:

7. Continental pooling – a facility could be created that bundles currency risks across the continent to give smaller markets the chance to diversify their economies. Bundling could also occur in projects with similar latitudes across different countries to create sector-specific investment vehicles. For example, having pools that combine solar projects with flood management systems may create a natural hedge - when one faces climate stress, the other becomes more valuable.
8. Technology enabled solutions – using machine learning to accurately price local currency risk potentially reducing risk premiums through improved data analysis. Further distributed ledger systems could be implemented for faster, cheaper cross border settlements of local currency transactions. Pilot DLT-based bond issuance like Abu Dhabi's HSBC partnership, to reduce settlement times and costs. Using digital currencies to settle intra-African trade in local currencies and having public dashboards to track FX exposures in real-time. Further, having currency corridors that widen or narrow based on commodity prices, inflation differentials, capital flow patterns and other economic fundamentals instead of having fixed exchange rates.
9. Solid guarantee structures – establish automatic payouts based on objective triggers (currency depreciation thresholds, economic indicators) to reduce administrative costs and have groups of similar borrowers i, e. Mutual Guarantee Societies: to cross-guarantee each other, spreading risks while reducing costs. Scale up public-private partnerships where governments cover 10-20% of losses, while DFIs (e.g., World Bank) provide reinsurance.
10. Risk management – multi-layered approach for example portfolio level for country states; weather-indexed hedging especially for countries that are heavily dependent on commodity exports allowing them to combine currency hedging with weather derivatives to address underlying economic volatility. Establishing early warning systems using predictive analytics to anticipate currency stress and trigger automatic support mechanisms before a crisis, stress tests and circuit breakers that activates during extreme market volatility for example, IFC's Managed Co-Lending Portfolio Program (MCP) to attract institutional investors.

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